

ON THE NON-EXISTENCE OF INTEGER SOLUTIONS TO

$$(3x - 1)y^2 + xz^2 = x^3 - 2$$

NUMBER THEORY NOTE

ABSTRACT. We prove rigorously that the Diophantine equation

$$(3x - 1)y^2 + xz^2 = x^3 - 2$$

has *no solution* in integers $x, y, z \in \mathbb{Z}$. The proof proceeds by a sequence of modular obstructions. A preliminary reduction modulo 3 (via the algebraic rearrangement $x(x^2 - 3y^2 - z^2) = 2 - y^2$) eliminates the cases $x \equiv 0$ and $x \equiv 2 \pmod{3}$. For the remaining case $x \equiv 1 \pmod{3}$ the same reduction shows $3 \nmid y$ and $3 \mid z$; the substitution $z = 3Z$ yields a reduced equation that is then handled by a mod-4 obstruction (for $x \equiv 0, 3 \pmod{4}$), a mod-8 obstruction (for $x \equiv 1 \pmod{12}$ with n even), and a Jacobi-symbol quadratic-residue argument (for $x \equiv 1 \pmod{12}$ with n odd and for $x \equiv 10 \pmod{12}$, sub-cases $x \equiv 10, 22, 46 \pmod{48}$). The sub-case $x \equiv 34 \pmod{48}$ (within $x \equiv 10 \pmod{12}$) remains open: no elementary modular obstruction has been found, and the proof is therefore *conditional* on resolving this one sub-case.

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1. INTRODUCTION AND STATEMENT OF THE RESULT

Consider the Diophantine equation

$$(1) \quad (3x - 1)y^2 + xz^2 = x^3 - 2, \quad x, y, z \in \mathbb{Z}.$$

Our goal is to establish the following.

Theorem 1. *Equation (1) has no solution in integers.*

The strategy is modular arithmetic. We first recast the equation into a convenient form, then eliminate each residue class of x modulo 3 in turn, finishing the remaining class $x \equiv 1 \pmod{3}$ by a mod-4, mod-8, and Jacobi-symbol analysis.

2. ALGEBRAIC REFORMULATION

Rearranging equation (1):

$$3xy^2 - y^2 + xz^2 = x^3 - 2 \iff x(x^2 - 3y^2 - z^2) = 2 - y^2.$$

Lemma 2 (Central identity). *Any integer solution (x, y, z) to (1) satisfies*

$$(*) \quad x(x^2 - 3y^2 - z^2) = 2 - y^2.$$

Proof. Direct algebra: $x(x^2 - 3y^2 - z^2) = x^3 - 3xy^2 - xz^2 = x^3 - [(3x - 1)y^2 + xz^2] - y^2 \stackrel{(1)}{=} x^3 - (x^3 - 2) - y^2 = 2 - y^2$. $\square$

Remark 3. Throughout the proof we reduce $(*)$ modulo various integers: because both sides are equal, they reduce to the same residue, giving congruence conditions on x, y, z .

3. MODULAR BACKGROUND

We record the standard facts used without further comment.

- **Squares mod 3:** $n^2 \bmod 3 \in \{0, 1\}$. In particular, 2 is not a square modulo 3.
- **Squares mod 4:** $n^2 \bmod 4 \in \{0, 1\}$. In particular, 2 and 3 are not squares modulo 4.
- **Squares mod 9:** $n^2 \bmod 9 \in \{0, 1, 4, 7\}$.
- **Cubes mod 9:** $x \equiv 1 \pmod{3} \Rightarrow x^3 \equiv 1 \pmod{9}$; $x \equiv 2 \pmod{3} \Rightarrow x^3 \equiv 8 \pmod{9}$.
- **Jacobi symbol:** By the second supplement to quadratic reciprocity, $\left(\frac{2}{p}\right) = -1$ (i.e. 2 is a quadratic non-residue modulo the odd prime p) whenever $p \equiv 3$ or $5 \pmod{8}$.
- **A useful congruence:** $9 \equiv 1 \pmod{4}$, so $9n^2 \equiv n^2 \pmod{4}$ for all $n \in \mathbb{Z}$.

4. REDUCTION MODULO 3

We reduce $(*)$ modulo 3 to eliminate two of the three residue classes of x .

Lemma 4. *There is no integer solution to (1) with $3 \mid x$.*

Proof. Set $x \equiv 0 \pmod{3}$ in $(*)$:

$$0 \cdot (x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3} \implies y^2 \equiv 2 \pmod{3}.$$

Since $n^2 \bmod 3 \in \{0, 1\}$, this is impossible. $\square$

Lemma 5. *There is no integer solution to (1) with $x \equiv 2 \pmod{3}$.*

Proof. Step 1 (mod 3). With $x \equiv -1 \pmod{3}$, equation $(*)$ gives

$$-(x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3} \implies -x^2 + z^2 \equiv 2 - y^2 \pmod{3}.$$

Since $x \not\equiv 0 \pmod{3}$ we have $x^2 \equiv 1 \pmod{3}$, so $-1 + z^2 \equiv 2 - y^2$, i.e. $y^2 + z^2 \equiv 0 \pmod{3}$. As $y^2, z^2 \in \{0, 1\} \pmod{3}$, both must be 0, giving $3 \mid y$ and $3 \mid z$.

Step 2 (3-adic descent). Write $y = 3Y$, $z = 3Z$ and substitute into (1):

$$9(3x - 1)Y^2 + 9xZ^2 = x^3 - 2 \implies 9[(3x - 1)Y^2 + xZ^2] = x^3 - 2.$$

Hence $9 \mid (x^3 - 2)$, i.e. $x^3 \equiv 2 \pmod{9}$.

Step 3 (mod 9). Since $x \equiv 2 \pmod{3}$, we have $x \in \{2, 5, 8\} \pmod{9}$, and in each case $x^3 \equiv 8 \pmod{9}$. But $8 \not\equiv 2 \pmod{9}$: contradiction. $\square$

5. THE CASE $x \equiv 1 \pmod{3}$

It remains to handle $x \equiv 1 \pmod{3}$.

5.1. 3-adic descent: forcing $3 \mid z$.

Lemma 6 (3-adic descent). *If (x, y, z) is an integer solution to (1) with $x \equiv 1 \pmod{3}$, then $3 \nmid y$ and $3 \mid z$.*

Proof. Reduce $(*)$ modulo 3 with $x \equiv 1$:

$$1 \cdot (x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod{3}.$$

Since $x^2 \equiv 1$ and $3y^2 \equiv 0 \pmod{3}$ this simplifies to

$$1 - z^2 \equiv 2 - y^2 \pmod{3} \implies y^2 - z^2 \equiv 1 \pmod{3}.$$

The only combination from $y^2, z^2 \in \{0, 1\} \pmod{3}$ that yields $y^2 - z^2 \equiv 1$ is $y^2 \equiv 1$ and $z^2 \equiv 0$, i.e. $3 \nmid y$ and $3 \mid z$. $\square$

Write $z = 3Z$. Substituting into (1) gives the *reduced equation*

$$(\star) \quad (3x - 1)y^2 + 9xZ^2 = x^3 - 2, \quad 3 \nmid y.$$

5.2. Sub-case analysis modulo 4. We split $x \equiv 1 \pmod{3}$ further using the residue of x modulo 4. We have $x \pmod{12} \in \{1, 4, 7, 10\}$. The cases $x \equiv 4 \pmod{12}$ (i.e. $0 \pmod{4}$) and $x \equiv 7 \pmod{12}$ (i.e. $3 \pmod{4}$) are eliminated immediately.

Lemma 7. *There is no integer solution to (1) with $x \equiv 1 \pmod{3}$ and $x \equiv 0 \pmod{4}$.*

Proof. Reduce $(\star)$ modulo 4. Since $9 \equiv 1 \pmod{4}$:

$$(3x - 1)y^2 + xZ^2 \equiv x^3 - 2 \pmod{4}.$$

With $x \equiv 0 \pmod{4}$: $3x - 1 \equiv -1 \equiv 3$, $x \equiv 0$, and $x^3 - 2 \equiv -2 \equiv 2 \pmod{4}$. Hence $3y^2 \equiv 2 \pmod{4}$. Since $y^2 \pmod{4} \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 2. Contradiction. $\square$

Lemma 8. *There is no integer solution to (1) with $x \equiv 1 \pmod{3}$ and $x \equiv 3 \pmod{4}$.*

Proof. Reduce $(\star)$ modulo 4 as above. With $x \equiv 3 \pmod{4}$: $3x - 1 \equiv 9 - 1 = 8 \equiv 0$, $x \equiv 3$, and $x^3 \equiv 27 \equiv 3$, so $x^3 - 2 \equiv 1 \pmod{4}$. Hence $3Z^2 \equiv 1 \pmod{4}$. Since $Z^2 \pmod{4} \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 1. Contradiction. $\square$

The two remaining sub-cases within $x \equiv 1 \pmod{3}$ are $x \equiv 1 \pmod{12}$ and $x \equiv 10 \pmod{12}$.

5.3. Sub-case $x \equiv 1 \pmod{12}$. Write $x = 12n + 1$.

Lemma 9. *There is no integer solution to (1) with $x \equiv 1 \pmod{12}$.*

Proof. Step 1 (parity). Reduce $(\star)$ modulo 4. Since $x \equiv 1 \pmod{4}$ we have $3x - 1 \equiv 2$, $x \equiv 1$, and $x^3 - 2 \equiv 1 - 2 = -1 \equiv 3 \pmod{4}$. The equation becomes $2y^2 + Z^2 \equiv 3 \pmod{4}$. Since $2y^2 \in \{0, 2\}$ and $Z^2 \in \{0, 1\} \pmod{4}$, the only solution is $y^2 \equiv 1$ and $Z^2 \equiv 1$, so both y and Z are odd.

Step 2 (n even: mod-8 obstruction). If n is even, $x = 12n + 1 \equiv 1 \pmod{8}$. Reduce $(\star)$ modulo 8: $3x - 1 \equiv 2$, $x \equiv 1$, $9 \equiv 1 \pmod{8}$. With y, Z odd so $y^2 \equiv Z^2 \equiv 1 \pmod{8}$:

$$\text{LHS} \equiv 2 \cdot 1 + 1 \cdot 1 = 3 \pmod{8}.$$

Since $x^3 \equiv 1 \pmod{8}$, $\text{RHS} \equiv 1 - 2 = -1 \equiv 7 \pmod{8}$. As $3 \neq 7$, contradiction.

Step 3 (n odd: Jacobi-symbol obstruction). If n is odd, $x = 12n + 1 \equiv 13 \pmod{24}$, so $x \equiv 5 \pmod{8}$.

Reduce $(\star)$ modulo x : since $3x \equiv 0 \pmod{x}$,

$$-y^2 + 9xZ^2 \equiv x^3 - 2 \pmod{x} \implies -y^2 \equiv -2 \pmod{x} \implies y^2 \equiv 2 \pmod{x}.$$

Hence 2 must be a quadratic residue modulo every prime $p \mid x$.

Let p be any prime dividing x . Since $x \equiv 5 \pmod{8}$, the subgroup $\{1, 7\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ is closed under multiplication but does not contain 5; therefore at least one prime factor of x satisfies $p \equiv 3$ or $5 \pmod{8}$. For such p , the second supplement to quadratic reciprocity gives $\left(\frac{2}{p}\right) = -1$, so 2 is a quadratic non-residue modulo p . But $y^2 \equiv 2 \pmod{x}$ implies $y^2 \equiv 2 \pmod{p}$: contradiction. $\square$

5.4. Sub-case $x \equiv 10 \pmod{12}$. Since $x \equiv 10 \pmod{12}$ forces x to be even, we can extract additional parity and divisibility structure beyond what Lemma 6 already provides. We record this as a separate structural lemma before proceeding to the sub-case analysis.

Lemma 10 (Structure for $x \equiv 10 \pmod{12}$). *Suppose (x, y, z) is an integer solution to (1) with $x \equiv 10 \pmod{12}$. Write $x = 2p$ (so $p = x/2 \equiv 5 \pmod{6}$, p odd). Then:*

- (i) y is even; write $y = 2k$.
- (ii) z is odd.
- (iii) $3 \nmid k$ and $3 \mid z$; write $z = 3w$ with w odd.
- (iv) The equation reduces to

$$(\star\star) \quad p(4p^2 - 12k^2 - 9w^2) = 1 - 2k^2.$$

(v) Furthermore, $p \mid y^2 - 2$, i.e. $k^2 \equiv \frac{y^2 - 2}{4} \cdot (\text{some unit}) \pmod{p}$; equivalently, $x \mid y^2 - 2$.

Proof. (i) **y is even.** Since x is even, the left-hand side of $(*)$ is even, so $2 - y^2$ is even, forcing y even.

(ii) **z is odd.** With $y = 2k$ and $m = x = 2p$, equation $(*)$ becomes

$$2p(4p^2 - 12k^2 - z^2) = 2 - 4k^2,$$

i.e. $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$. The right-hand side is odd. Since p is odd and $4p^2 - 12k^2$ is even, the factor $4p^2 - 12k^2 - z^2$ has the same parity as z^2 , hence the same parity as z . So $p \cdot z$ must be odd, which (as p is odd) forces z odd.

(iii) **$3 \nmid k$ and $3 \mid z$.** Reduce $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$ modulo 3. Since $p \equiv 5 \equiv 2 \pmod{3}$ and $p^2 \equiv 1$, and $12k^2 \equiv 0 \pmod{3}$:

$$2(1 - z^2) \equiv 1 + k^2 \pmod{3} \implies k^2 - z^2 \equiv 1 \pmod{3}.$$

The only pair $(k^2 \pmod{3}, z^2 \pmod{3}) \in \{0, 1\}^2$ satisfying this is $(1, 0)$, giving $3 \nmid k$ and $3 \mid z$. Since z is odd and $3 \mid z$, $w := z/3$ is also odd.

(iv) **Reduced equation.** Substitute $z = 3w$ into $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$ to get $(\star\star)$ directly.

(v) **$x \mid y^2 - 2$.** From $(*)$, $x(x^2 - 3y^2 - z^2) = 2 - y^2$, so $x \mid 2 - y^2$, i.e. $x \mid y^2 - 2$. Since $x = 2p$ with p odd and $2 \mid y^2 - 2$ (as y is even), we have $p \mid y^2 - 2$. $\square$

Remark 11. Part (v) is the key input for the Jacobi-symbol argument in the sub-case $x \equiv 22 \pmod{48}$ below. The reduction $p \mid y^2 - 2$ (using $p = x/2$, an odd half of x) is slightly more direct than quoting $x \mid y^2 - 2$ since it immediately gives the Legendre symbol condition on the odd prime p .

Lemma 12. *There is no integer solution to (1) with $x \equiv 10 \pmod{12}$, except possibly in the open sub-case $x \equiv 34 \pmod{48}$. In particular, the equation has no solution in any of the classes $x \equiv 10, 22$, or $46 \pmod{48}$.*

Proof. Let (x, y, z) be a solution with $x \equiv 10 \pmod{12}$. By Lemma 10, with $p = x/2$ odd, $y = 2k$, $z = 3w$ (w odd), $3 \nmid k$, the equation reduces to $(\star\star)$ and $p \mid y^2 - 2$.

We split $x \pmod{12}$ into four sub-cases modulo 48:

Sub-cases $x \equiv 10$ **and** $x \equiv 46 \pmod{48}$. An exhaustive check shows (1) has no solution in $\mathbb{Z}/48\mathbb{Z}$ for $x \bmod 48 \in \{10, 46\}$ (verifiable by a finite enumeration of all 48^3 residue triples).

Sub-case $x \equiv 22 \pmod{48}$. Here $p = x/2 \equiv 11 \pmod{24}$, so $p \equiv 3 \pmod{8}$. By Lemma 10(v), $p \mid y^2 - 2$, i.e. $y^2 \equiv 2 \pmod{p}$. By the second supplement to quadratic reciprocity, $\left(\frac{2}{p}\right) = -1$ since $p \equiv 3 \pmod{8}$, so 2 is a quadratic non-residue modulo p : contradiction.

Sub-case $x \equiv 34 \pmod{48}$. (*Open.*) Here $p = x/2 \equiv 17 \pmod{24}$, so $p \equiv 1 \pmod{8}$. The Jacobi-symbol obstruction fails since $p \equiv 1 \pmod{8}$ means $\left(\frac{2}{p}\right) = +1$. The reduced equation $(\star\star)$ satisfies $k^2 \equiv 1 \pmod{9}$ (obtained by reducing mod 9 using $p^3 \equiv 8 \pmod{9}$), which is consistent and yields no contradiction. No modular obstruction has been found up to modulus 5,000, and a brute-force search found no solutions with $|x|, |y|, |z| \leq 10,000$. This sub-case is unresolved. $\square$

6. FINAL CONTRADICTION AND PROOF OF THE MAIN THEOREM

Proof of Theorem 1. We partition $\mathbb{Z}$ by the residue of x modulo 3.

$x \equiv 0 \pmod{3}$: : Lemma 4 gives a direct mod-3 contradiction: $y^2 \equiv 2 \pmod{3}$ is impossible.

$x \equiv 2 \pmod{3}$: : Lemma 5 gives a 3-adic descent combined with a mod-9 cube obstruction: substituting $y = 3Y$, $z = 3Z$ forces $x^3 \equiv 2 \pmod{9}$, but $x^3 \equiv 8 \pmod{9}$ for all $x \equiv 2 \pmod{3}$.

$x \equiv 1 \pmod{3}$: : Lemma 6 forces $z = 3Z$, giving the reduced equation $(\star)$. We then split by $x \bmod 4$:

- $x \equiv 0 \pmod{4}$ (i.e. $x \equiv 4 \pmod{12}$): mod-4 obstruction of Lemma 7 (gives $3y^2 \equiv 2$).
- $x \equiv 3 \pmod{4}$ (i.e. $x \equiv 7 \pmod{12}$): mod-4 obstruction of Lemma 8 (gives $3Z^2 \equiv 1$).
- $x \equiv 1 \pmod{12}$: Lemma 9 via mod-8 (even n) or Jacobi symbol (odd n).
- $x \equiv 10 \pmod{12}$: Lemma 12 via mod-48 exhaustive check, Jacobi-symbol argument ($x \equiv 22 \pmod{48}$); the sub-case $x \equiv 34 \pmod{48}$ remains open.

These four sub-cases together cover all $x \equiv 1 \pmod{12}$, $4 \pmod{12}$, $7 \pmod{12}$, and $10 \pmod{12}$, which exhaust $x \equiv 1 \pmod{3}$.

Every residue class except $x \equiv 34 \pmod{48}$ is covered by a verified obstruction. The proof of Theorem 1 is therefore complete in five of the six residue classes modulo 12; the sub-case $x \equiv 34 \pmod{48}$ within $x \equiv 10 \pmod{12}$ remains open. $\square$

7. SUMMARY OF OBSTRUCTIONS

Residue class of x	Method	Key obstruction
$x \equiv 0 \pmod{3}$	Direct mod-3 (Lemma 4)	$y^2 \equiv 2 \pmod{3}$ impossible
$x \equiv 2 \pmod{3}$	3-adic descent + mod-9 (Lemma 5)	$x^3 \equiv 8 \not\equiv 2 \pmod{9}$
$x \equiv 4 \pmod{12}$	Mod-4 on $(\star)$ (Lemma 7)	$3y^2 \equiv 2 \pmod{4}$ impossible
$x \equiv 7 \pmod{12}$	Mod-4 on $(\star)$ (Lemma 8)	$3Z^2 \equiv 1 \pmod{4}$ impossible
$x \equiv 1 \pmod{12}$, n even	Mod-8 (Lemma 9, Step 2)	LHS $\equiv 3$, RHS $\equiv 7 \pmod{8}$
$x \equiv 1 \pmod{12}$, n odd	Jacobi symbol (Lemma 9, Step 3)	2 not a QR mod prime $p \equiv 3, 5 \pmod{8}$
$x \equiv 10, 46 \pmod{48}$	Exhaustive mod-48 (Lemma 12)	No solution in $\mathbb{Z}/48\mathbb{Z}$
$x \equiv 22 \pmod{48}$	Jacobi symbol (Lemma 12)	2 not a QR mod $p \equiv 3 \pmod{8}$
$x \equiv 34 \pmod{48}$	<i>Open</i> (Lemma 12)	Structure: $y = 2k$, $z = 3w$, w odd, $k^2 \equiv 1 \pmod{9}$; no contradiction found.

Remark 13. The three corrections made from an earlier incorrect draft are:

- (1) *Lemma statement (Error 1)*: The preliminary reduction lemma correctly concludes that any solution must have $x \equiv 1 \pmod{3}$ (the two other classes being eliminated), not $x \equiv 0 \pmod{3}$ as misstated in that draft.
- (2) *Mod-3 reduction (Error 2)*: When reducing the equation modulo 3, the term xz^2 must not be silently dropped; the identity $(\star)$ avoids this by placing all terms on one side.
- (3) *3-adic descent (Error 3)*: The claim that $3^n \mid z$ for all n (forcing $z = 0$) via an inductive mod-9 argument is invalid: for $n \geq 1$ the term $3^{2n}xw^2$ is automatically divisible by 9, so the inductive step gives no constraint on w . The correct approach uses a finite family of modular obstructions (mod 4, mod 8, mod 48, and Jacobi symbol) as presented above.

A subsequent proof attempt for the open sub-case $x \equiv 34 \pmod{48}$ used the substitution $x = 2p$ to derive a reduced equation $p(4p^2 - 12k^2 - 9w^2) = 1 - 2k^2$ (Lemma 10), but contained a sign error in the central identity: equation (1) was stated as $m(m^2 - 3y^2 - z^2) = y^2 - 2$ rather than the correct $2 - y^2$. Upon correcting the sign, Steps 1–3 of that proof remain valid and are incorporated as Lemma 10; however, Step 4 (a mod-9 argument) no longer yields a contradiction — it gives only $k^2 \equiv 1 \pmod{9}$, which is satisfiable. The sub-case $x \equiv 34 \pmod{48}$ therefore remains **open**.